CHAPTER

Prelude to the Process Chapters

5

Highlights

- How UX processes, methods, and techniques intertwine over time.
- The UX design studio.
- Project commission: How does a project start?
- Key UX roles.
- The Middleburg University Ticket Transaction Service and the new Ticket Kiosk System: Our running examples for the process chapters.
- The product concept statement:
  - How to write a clear and concise product concept statement.

5.1 INTRODUCTION

This minichapter is an introduction to the UX process chapters. It contains a mix of a few topics prefatory to the process chapters.

5.2 INTERTWINING OF PROCESSES, METHODS, AND TECHNIQUES

Although we present the UX lifecycle activities in separate chapters to focus on learning about each one at a time, the reality of UX practice is that the UX process is a fabric of interwoven lifecycle activities, methods, and techniques. UX design activities are not performed in a nice orderly sequence but are intertwined within the process, even occurring simultaneously.

Lifecycle activity

The high-level things you do during a UX lifecycle, including understanding user needs, designing UX solutions, prototyping candidate designs, and evaluating those designs (Section 2.2.2).

Examples of Intertwining

As an example of intertwining, consider the important UX design activity of prototyping. For starters, prototyping is featured as a major lifecycle activity in a chapter of its own. But prototyping is also an indispensable UX technique that appears in many different guises in many different places. A prototype is a tangible manifestation of any idea that needs to be evaluated. Prototyping in some form can appear in almost any method for performing almost any activity.

In addition to being used as the platform for UX evaluation (Part 5), multiple low-fidelity rapid prototypes, called sketches in this context, are used to explore competing design ideas in ideation (a kind of brainstorming, Chapter 14).

As a concrete example, a wireframe is an important kind of prototype. Quick and dirty wireframes are used as sketches for exploring design ideas as part of early design. Finished and detailed wireframes are used to convey design ideas to stakeholders and for implementation in Chapters 17 and 18 on designing the interaction and designing for emotional impact.

Wireframe

A visual template of a screen or webpage design in the interaction perspective, comprised of lines and outlines (hence the name "wire frame"). A skeletal representation of the layout of interaction objects such as tabs, menus, buttons, dialogue boxes, displays, and navigational elements (Section 17.5).

5.2.1 Activity Timing

Another area where practice strays from the "pure" activity description is the timing of how things happen. For example, in our usage research data elicitation discussion (observation and interviewing of users, Chapter 7), it might seem that all user visits happen in one contiguous time period. But in practice that is rarely the case. The data elicitation activity can occur on and off over a period of time because of a variety of project constraints, including scheduling, user availability, and concurrent and unrelated projects happening simultaneously. In our experience, we have never been able to do all data elicitation in one go even if we had a large team of researchers and designers. We could never schedule them like that. So we would start with the first user, gather usage data, and start early modeling and synthesis and even some design while we wait for the next user visit. After that next visit, we would update the models and the design ideas with the new data as necessary.

To a lesser extent but still not uncommon are situations where we would even have some early designs and prototypes done that we would take to users that were scheduled further out on the calendar. If we tried to account for that in the elicitation discussion, we would have had to cover the material on early modeling, design ideation, sketching, and early wireframing in multiple and overlapping discussions. It would have resulted in a mashup that was difficult to sort out by you, the reader.

The practice of UX evaluation presents another example of how timing can vary. For example, we present design production in isolation as a chapter in the book part on design. UX evaluation comes in later chapters, but in fact by the time design production is done, most formative evaluation (evaluation for the purpose of refining the design) is usually done, too. And before wireframing is done, most other kinds of prototyping (building a preliminary version, Chapter 20) are also done. A true-to-life description of this would bring much of the evaluation and prototyping discussion upfront before design production is complete. Here is a typical sequence of activities we have seen in practice:

- Data elicitation with few early users.
- Analysis and modeling of data collected.
- Ideation and sketching for generative design based on analysis and emerging models.
- Early wireframes or prototypes of emerging designs.
- Data elicitation from more users:
  - Show users emerging design ideas and wireframes at the end of elicitation sessions for feedback.
- Update usage data models and designs with new insights.
- Increase fidelity of wireframe details as confidence on emerging designs grows.
- Evaluate with more users using more rigorous methods (e.g., in a UX lab).
- Update usage data models and designs with evaluation findings.
- Design production and hand off to the software engineering team.

5.2.2 Can We Describe It that Way in a Book?

If we try to account for these parallel and concurrent streams of activities in a stream-of-occurrence style in this book, we run the risk of repetition. Even if we could pull that off, the result would be a little of this and a little of that, with no complete description of anything in one place. That makes a confusing jumble for the reader who is trying to understand each part of the process.

As an analogy for the reader, it would be like someone trying to learn the process by observing a UX professional—you see the process unfold in overlapping bits. You have to deduce how to piece it together. If something changes, you have to update your understanding until you eventually have the big picture that extensive experience brings.

To resolve this in favor of clarity for the reader, our process chapters are organized mainly by UX design activity. But we also try to indicate how these activities actually happen in the field. To this end, we also use some incremental disclosure. For example, in Chapter 7 on data elicitation for usage research, we have a section on early data modeling where we reveal just enough to get started on modeling in parallel with data elicitation. But we don't say everything about modeling here because that would be a distraction from data elicitation.

Wireframe

A visual template of a screen or webpage design in the interaction perspective, comprised of lines and outlines (hence the name "wire frame"). A skeletal representation of the layout of interaction objects such as tabs, menus, buttons, dialogue boxes, displays, and navigational elements (Section 17.5).

Formative evaluation

A family of diagnostic UX evaluation methods using mostly qualitative data collection with the objective to form a design, that is, for finding and fixing UX problems and thereby refining the design (Section 21.1.5).

Similarly, we introduce some early wireframing in the chapter on generative design (UX design creation, Chapter 14) but reserve the full treatment of wireframing for the prototyping chapter (Chapter 20). We also have a little bit about early informal evaluation to show how we critique these early wireframes.

5.2.3 Readers Need a "Pure" Description of Each Lifecycle Activity

For clarity, we feel the reader must start with a more or less "pure" description of each major UX design lifecycle activity separately, without the distraction of how they are interleaved. That doesn't follow the UX principle of giving information to the user/reader just as needed, but it is the only approach that:

- Facilitates immersion by focusing the reader on a single topic at a time.
- Limits repetition in the narrative by covering a topic only once.
- Effectively helps readers use this book as a reference while doing a particular UX activity.

Once armed with these modules of knowledge, we hope the student or practitioner can adopt, adapt, and interleave the appropriate choices and combinations of methods and techniques as needed in any particular design situation or project.

5.3 A DEDICATED UX DESIGN STUDIO, AN ESSENTIAL TOOL FOR TEAMWORK

Before we get started with usage research data elicitation, this is a good time to talk about an indispensable tool for every UX group: *the UX design studio.*

5.3.1 Why You Need a UX Design Studio

A UX design studio is:

- A home for your team.
- A place for immersion.
- A shared workspace for colocation and constant collaboration.
- A place for externalizing the state of the design by posting all your work for discussion and brainstorming.

As Bødker and Buur (2002) urge, every UX group needs to set aside a physical work space, a design studio, as a home base for the team to meet and do ideation, individual work, design collaboration, and other group work.

Immersion

A form of deep thought and analysis of the problem at hand—to "live" within the context of a problem and to make connections among the different aspects of it (Section 2.4.7).

The UX studio is where you immerse your team in the artifacts of analysis and design. When you walk in the room and close the door, you are in the world of the design and the rest of the world and its distractions disappear. This means you need your whole team colocated enough so you can all meet in the UX studio at almost any time (Brown, 2008, p. 87).

5.3.2 What You Need in Your UX Design Studio

A good design studio will have:

- Comfortable seating.
- A conference table.
- Work tables to lay out prototypes and other design artifacts.
- White boards for group sketching.
- Ample wall space for hanging posters, drawings, diagrams, and design sketches.
- A door that can be closed.
- Shared displays on which team members can show work artifacts from their laptops.

5.3.3 Dedicated Space

Your UX design studio space needs to be a place that:

- Is constantly available without the need for reservations.
- Allows all artifacts to remain displayed from one work session to another.
- Meets the unpredictable need for frequent and immediate communication.

5.3.4 The Virginia Tech Industrial Designs Studio: The Kiva

In Fig. 5-1, we show an example of a collaborative ideation and design studio, called the Kiva, in the Virginia Tech Department of Industrial Design. The Kiva was originally designed and developed by Maya Design in Pittsburgh, and is used at Virginia Tech with their permission.

The Kiva is a cylindrical space in which designers can brainstorm and sketch in isolation from outside distractions. The space inside is large enough for seating and work tables. The inner surface is a painted metallic skin that serves as an enveloping whiteboard and can also hold magnetic “push pins.” The large-screen display on the outside can be used for announcements, including group scheduling for the work space.

In Fig. 5-2, we show immersion and collaboration within individual and group workspaces for colocation of designers, just outside the Kiva.

Fig. 5-1
The Virginia Tech ideation and design studio, the "Kiva" (photo courtesy of Akshay Sharma of the Virginia Tech Department of Industrial Design).

Fig. 5-2
Individual and group designer workspaces (photos courtesy of Akshay Sharma of the Virginia Tech Department of Industrial Design).

5.4 THE PROJECT COMMISSION: HOW DOES A PROJECT START?

All projects have some kind of starting point. Projects are spawned by any of a number of forces, including marketing strategy, management edict, a product sponsor, a system client, inventive ideas, perceived needs to improve an existing product or system, and so on.

The commission to undertake a system-oriented project rarely comes from within the UX design or software engineering (SE) group, but instead as a mandate from upper management. The notion of a new product can hatch in the form of a business brief, often from marketing within an established product-oriented organization.

For the project team, things begin with some kind of project commission document or statement that is a proposal for creating the project. Depending on the organizational situation, the project commission document can include boiler plate items such as project duration, budget, management plan, personnel roles, etc.

A project commission document is likely to be very high level and possibly ill defined. It becomes the first responsibility of the UX design team, especially the product owner (just below in Section 5.4.1.5), to fill in the gaps during the product definition stage and firm up the vision in the product concept statement.

Product concept statement

Concise mission statement for the project team, a vision statement or a mandate, an explanation of the product to stakeholders and outsiders. Internally, it is a yardstick to help set focus and scope for development. The audience is very broad, including high-level management, marketing, stockholders, and even the general public (Section 5.6).

5.4.1 Key UX Team Roles from the Start

5.4.1.1 Usage researcher

This is a person who does usage research activities such as data elicitation, data analysis and modeling, and UX requirements specifications.

5.4.1.2 UX designer

This is a person on the UX team who does UX design, such as design creation, conceptual design, and/or design production for user interaction.

5.4.1.3 Graphic or visual designer

A graphic designer is responsible for visual communication, branding, and style, eventually in pixel-perfect visual "comps."

5.4.1.4 UX analyst

This is a person on the UX team who does UX evaluation. Often, in small teams, the same person may perform the duties of usage researcher and UX analyst.

5.4.1.5 Product owner

Some UX teams have either a product owner, a product manager, or both. The definitions of the roles of product owner and product manager can vary between project teams. The product owner is more likely to be associated with the UX team. Many companies don't even have a product manager, whose role is mainly about setting up meetings and scheduling.

Typically, the product owner:

- Is close to users.
- Knows the product context within the client's domain, business strategy, and competition.

User story

A short narrative describing a feature or capability needed by a user in a specific work role and a rationale for why it is needed, used as an agile UX design "requirement" (Section 10.2.2).

Persona

A narrative description of a specific design target of a UX design for a user in one work role, a hypothetical but specific "character" in a specific work role (Section 9.4).

Enterprise system

A large information system used within an organization, typical of those developed and used within IT departments in organizations (Sections 3.2.2.4 and 3.4.2).

- Is responsible for the success of the product and for pursuing business goals.
- Keeps the product on track with the overall vision.
- Is in charge of writing/creating:
  - The product concept statement.
  - User stories.
  - User personas.
- Works with SE in planning implementation sprints and managing prioritized backlog of agile user stories (Chapter 29).

5.5 THE MIDDLEBURG UNIVERSITY TICKET TRANSACTION SERVICE AND THE NEW TICKET KIOSK SYSTEM

As a running example to illustrate the ideas in the text, we introduce the Middleburg University Ticket Transaction Service (MUTTS), a hypothetical public ticket sales system for selling tickets for entertainment and other events. This information about MUTTS is the kind of thing you discover in data elicitation, the subject of Chapter 7. This project is a good example of a combined product perspective and enterprise system perspective—it's a system but it's used much like a product by the general public.

5.5.1 The Existing System: The Middleburg University Ticket Transaction Service (MUTTS)

Middleburg, a small town in middle America, is home to Middleburg University, a state university that operates a service called the Middleburg University Ticket Transaction Service (MUTTS). MUTTS has operated successfully for several years as a central campus ticket office where people buy tickets for entertainment events, including concerts, plays, and special presentations by public speakers. Through this office, MUTTS makes arrangements with event sponsors and sells tickets to various customers. There is considerable interest in improving and expanding MUTTS.

The current business process suffers from numerous drawbacks:

- Until recently, all customers had to go to one location to buy tickets in person.
- MUTTS has now partnered with Tickets4ever.com as a national online ticket distribution platform. However, Tickets4ever.com suffers from low reliability and has a reputation for a poor user experience.

- Current operation of MUTTS involves multiple systems that do not work together very well.
- The ability to rapidly hire temporary ticket sellers to meet periodic high demand is hampered by university and state hiring policies.

5.5.1.1 Organizational context of the existing system

The desire to expand the business coincides with a number of other dynamics currently affecting MUTTS and Middleburg University.

- The supervisor of MUTTS wishes to expand revenue-generating activities.
- There is a general consensus that the new system should provide a much improved user experience in several areas.
- To leverage their increasing national academic and athletic prominence, the university is seeking a comprehensive customized solution that includes the integration of tickets for athletic events. Currently, tickets to athletic events are managed by an entirely different department.
- By including tickets for athletic events that generate significant revenue, MUTTS will have access to resources to support other aspects of their expansion.
- The university is undergoing a strategic initiative for unified branding across all its departments and activities. The university administration is receptive to creative design solutions for MUTTS to support this branding effort.

5.5.2 The Proposed New System: The Ticket Kiosk System

The working name for the new system is the Ticket Kiosk System, which was commissioned in a business brief by the university administration and MUTTS management. A high-level decision has been made to mandate a ticket kiosk as the core of this project. As designers, we need to keep an open mind and be prepared to steer the project toward a better solution, if one arises upon immersion in the problem space.

Early on, the scope of the project must be addressed in the business brief as a road map for all that is to follow. For example, does the design cover kiosk maintenance, payment arrangements, arrangement of agreements for purchase of tickets to outside organizations, and so on? These issues reflect other work roles in the ecology of the system, each of which, if included in the project, will have its own set of tasks to be designed for, especially in the interaction design.

Ecology

In the setting of UX design, the ecology is the entire set of surrounding parts of the world, including networks, other users, devices, and information structures, with which a user, product, or system interacts (Section 16.2.1).

5.5.3 Rationale

MUTTS wants to expand its scope and add more locations, but it is expensive to rent space in business buildings around town and the kind of small space it needs is rarely available. Therefore, the administrators of MUTTS and the Middleburg University administration have decided to switch the business from a ticket window to kiosks, which can be placed in many more locations across campus and around town.

Middleburg has a reliable and well-used public transportation system operated by Middleburg Transit, Inc. There are several bus stops, including at the library and the shopping mall, where buses come and go every few minutes with good-sized crowds getting on and off. Most such locations have space for a kiosk at a reasonable leasing fee.

Management expects an increase in sales due to the increased availability (kiosks in many places) and increased accessibility (open continuously). Also, there will be cost savings in that a kiosk requires no personnel at the sales outlets.

5.6 THE PRODUCT CONCEPT STATEMENT

It's the job of the product owner and the UX team to crystallize the product concept in a clear mission statement addressed to all stakeholders (Note: depending on the whether you are designing a commercial product or an enterprise system, you will have a product concept statement or a system concept statement. For simplicity, we use the term "product concept statement" as a more general term).

Stephenie Landry, vice president of Amazon Prime Now, starts a project by writing the product concept statement as a press release, working backward from there to deduce high-level needs and requirements and design. In the product concept statement for Amazon Prime Now, she promised a user experience that would be "magical" (Landry, 2016).

Tony Fadell, creator of the Apple iPod (Moore, 2017), who says product design and marketing are closely related, also creates a press release as one of the first steps in designing a new product.

The nature of a product concept statement. A product concept statement is:

- Concise, typically 100–150 words in length.
- A mission statement for the project team, a vision statement, or a mandate.
- A way to explain the product to stakeholders and outsiders.
- A yardstick to help set focus and scope for development internally.
- Not easy to write (especially a good one).

Because of the need for the statement to be short and concise, every word is important. A concept statement is not just written; it is iterated and refined to make it as clear and specific as possible.

- The audience for the product concept statement is very broad, including high-level management, marketing, the board of directors, stockholders, and even the general public.
- Refine and update as necessary as you proceed through the lifecycle activities.
- Post your product concept statement in your design studio as soon as possible. It will guide all UX design activities.

5.6.1 What's in a Product Concept Statement?

An effective product concept statement answers at least the following questions:

- What is the product or system name?
- Who are the users?
- How will they use it?
- System: What problem(s) will the system solve (broadly including business objectives)?
- Product: What are the major attractions to, or distinguishing features of, this product?
- What is the design vision and what are the emotional impact goals? In other words, what experience will the product or system provide to the user?

Example: Product Concept Statement for the Ticket Kiosk System

The TKS will replace MUTTS, the old ticket retail system, by providing distributed kiosk service to the general public. This service includes access to comprehensive event information and the capability to rapidly purchase tickets for local events such as concerts, movies, and the performing arts.

The new system includes a significant expansion of scope to include ticket distribution for the entire MU athletic program. Transportation tickets will also be available, along with directions and parking information for specific venues. Compared to conventional ticket outlets, the Ticket Kiosk System will offer increased (24/7) access, reducing waiting time, and far more extensive information about events. A focus on innovative design will enhance the MU public profile while fostering the spirit of the MU community and offering customers a quality experience. (128 words)

Iteration and refinement. Your product concept statement will evolve as you proceed with the project. For example, "far more extensive information about events" can be made more specific by saying "extensive information including images, movie trailers, and reviews of events." Also, we did not mention security and privacy, but these important concerns are later pointed out by potential users. Similarly, the point about "focus on innovative design" can be made more specific by saying "the goal of innovative design is to reinvent the experience of interacting with a kiosk by providing an engaging and enjoyable transaction experience."

Concept statements for large systems. For very large systems, you might need a system concept statement for each major subsystem, if each subsystem has its own goals and mission statement.

5.6.2 Introduction to Process-Related Exercises

How to approach these exercises: The exercises are for learning, not for producing a product, so you don't have to complete every detail if you think you have gotten what you need out of it. You should be able to learn most of what you can get from most exercises in an hour or so. In the case of a team within a classroom setting, this means that you can do the exercises as in-class activities, breaking out into teams and working in parallel, and possibly finishing the exercise as homework before the next class. This has the advantages of working next to other teams with similar goals and problems and of having an instructor present who can move among teams as a consultant and mentor. We recommend that student team deliverables be prepared in summary form for presentation to the rest of the class, if you have time, so that each team can learn from the others.

Choosing a product or system for these exercises: Choosing a product or system for these exercises should take some thought because you'll be using the same choice for most of the exercises throughout the process chapters.

Your choice of a target application system for the exercises should be gauged toward the goal of learning. That means choosing something the right size. Avoid applications that are too large or complex; choose something for which the semantics and functionality are relatively easy to understand. However, avoid systems that are too small or too simple because they may not support the process activities very well.

The criterion for selection here is that you will need to identify at least a half-dozen somewhat different kinds of user tasks. Look for something with an interesting ecology. You should also choose a system that has more than one class of user. For example, an e-commerce website for ordering merchandise will have users from the public doing the ordering and employee users processing the orders.

For practitioner teams in a real development organization, we recommend against using a real development project for these exercises. There is no sense in introducing the pressure to produce a real design and the risk of failure into this learning process.

Exercise 5.1: Product Concept Statement for a Product or System of Your Choice

Goal: Get practice in writing a concise product concept statement.
Activities: Write a product concept statement for a product or system of your choice. Iterate and polish it. The 150 or fewer words you write here will be among the most important words in the whole project, so write thoughtfully.
Deliverables: Your product concept statement.
Schedule: Given the simplicity of the domain, we expect you can get what you need from this exercise in about 30 minutes.

5.7 WELCOME TO THE PROCESS CHAPTERS

After the next chapter, the background chapter for Part 1, we move into Part 2 of the book, where we do usage research to understand needs and requirements. Part 3 is all about design, followed by Part 4, which is about prototyping. The process chapters are completed in Part 5, about UX evaluation.